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Restricting layers, flow paths, and correlation between duration of soil saturation and soil morphological features along a hillslope with an altered soil water regime in western Victoria

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Abstract

This paper is the second of two describing how soil macromorphological and chemical data can be combined with soil hydrological data to distinguish between, and to quantify, past and present hydrological processes relevant to waterlogging and dryland salinity. The first paper provides a methodological framework for the study, and describes the initial interpretation of the macromorphological features of the toposequence studied, using the soil feature–system–domain grouping method. This second paper deals with the added value of extensive piezometric and other hydrological observations relative to soil macromorphological studies, and with quantitative relationships between soil colour and duration of waterlogging.

As with the first paper, this paper focuses on a soil toposequence at Gatum on the eastern Dundas Tableland in western Victoria. For the broad crest with yellow gradational soils or Dermosols (Plinthoxeralfs), the hydrological data confirmed the conclusion from macromorphological observations that:

- (1) There are three levels at which downward flow of water is restricted: at the top of the largely unaltered mottled zone, that is at 0.8–1.0 m depth, well below the top of the yellow Bt1-horizon (fresh water); at the top of the pallid zone, at about 3–3.5 m depth (fresh water); and on top of the unweathered ignimbrite (saline water).
- (2) Below about 30 mm depth, down at least as far as 2.0 m and sometimes even more, the major pathways for downward movement of water are indeed root channels, with or without live roots, and not interpedal cracks. The 3-dimensional spacing of the rootholes above and through the less-permeable mottled zone, and the (horizontal) conductivity, storage capacity, and thickness of overlying horizons determine the extent of ponding, runoff, and deep infiltration taking place.

In addition, the hydrological observations showed that:

- (3) If rainfall is regular, well-spaced, and not excessive, ponding is unlikely to take place even where there is a rainfall surplus and there are layers restricting downward flow of water.

Furthermore, along the toposequence with yellow duplex soils or Dermosols, Chromosols, and Hydrosols,

- (4) There were generally good correlations between duration of saturation at the bottom of the E-horizon and colour aspects of the E-horizon (value and chroma of the matrix) and of the B2-horizon (hue, value, and chroma of the matrix).

Based on these findings additional suggestions are made for improving identification and management of water logging and salinisation processes.

Additional keywords: soil colour, redoximorphic features, waterlogging, perched aquifers, Plinthoxeralf-Natrixeralf-Plinthoxeralf-Natraqualf hydro-toposequence.

Introduction

Waterlogging or saturation of topsoils is a major constraint to agricultural production in southern Australia (e.g. McFarlane and Cox 1992; Cox *et al.* 1996; Fitzpatrick *et al.* 1996). Soil saturation is often associated with dryland salinity, which also causes large losses in

agricultural production (e.g. Robertson 1993). To understand both phenomena, and to find efficient solutions to them, a good understanding of local soil hydrology is essential. Unfortunately, soil hydrological monitoring is time-consuming and expensive, even under favourable circumstances. In addition, one never knows beforehand whether the rainfall during the monitoring period will be high, low, or average. One way around this problem is to determine relationships between soil macromorphology and soil hydrology that can be extrapolated in both time and space. Soil macromorphological characteristics generally do not change very fast, and are often a useful reflection of average soil hydrological conditions at a particular site over a number of years. At other sites, with similar geology, soils, and climate conditions, relationships between soil macromorphology and soil hydrology are likely to be very similar. This makes it possible to establish quantitative as well as qualitative relationships between soil macromorphology and soil hydrology at one site, and to use these relationships at another similar site to predict, qualitatively and quantitatively, soil hydrological conditions from soil macromorphological observations made during a single visit.

Several approaches have been used to establish relationships between soils and soil water flow. There have been many attempts to relate the colour of various soil features to duration of soil saturation. Much of the relevant North American work, and some European work, is summarised in Bigham and Ciolkosz (1993). The traditional approach describes variation of soil morphological features in soil horizons, such as A-horizon thickness and depth to mottles in B-horizons. That variation is then related to differences in slope position and in duration of saturation (e.g. Daniels *et al.* 1971; Franzmeier *et al.* 1983; Vepraskas and Wilding 1983a, 1983b; Coventry and Williams 1984; Blume and Schlichting 1985; Mokma and Cremeens 1986, 1991; Evans and Franzmeier 1988; Vepraskas *et al.* 1994). A similar but more systematic procedure is to use the structural approach (Boulet *et al.* 1982; Fritsch *et al.* 1992) to group soil morphological features across horizons and down toposequences into soil systems and soil domains. These domains and systems are used to illustrate the main interactions between water flow systems and soil processes (Fritsch and Fitzpatrick 1994). In addition, terrain-modeling techniques based on mathematical characterisation of local topography have been used to characterise surface hydrology (e.g. Moore *et al.* 1991) and predict surface soil properties (e.g. O'Loughlin 1986; Moore *et al.* 1993). However, on nearly level landscapes, such as in semi-arid north Queensland, subsurface restrictive layers control flow processes and are the most important factors governing soil differences between the red, yellow, and grey earths (Coventry and Williams 1984). Clearly, important differences in soil properties may also occur where topographic differences are very small and not necessarily detected by terrain modelling techniques.

In this study we have hypothesised that the variability in soil water duration and flow across a hillslope is influenced by subsurface layering as much as by landscape topography. We present the results of fieldwork involving a toposequence of Typic Plinthoxeralfs–Aquic Natrixeralfs–Typic Plinthoxeralfs–Typic Natraqualfs on the eastern Dundas Tableland in western Victoria. Both waterlogging and dryland salinity form considerable limitations to agricultural production in that area. The main objective of the project was to find economically attractive ways to reduce groundwater recharge under pastures, primarily by increasing pasture production and thus evapotranspiration. However, it was found that the hydrological behaviour of the soils initially examined did not conform to conventional model for such texture-contrast soils in Australia, i.e. ponding of water on top of the B-horizon, even though they did seem to exhibit the normal macromorphological characteristics. This led to a more detailed study of the relationships, in that landscape,

between geology, soil profile macromorphology, soil mineralogy, and soil hydrology. During this study, comparisons were also made of the advantages and disadvantages of each of these disciplines in improving understanding of local, landscape-wide, soil-related processes. The results of the geological and macromorphological interpretations, using the above-mentioned structural approach, and a summary of the hydrological findings, are presented in Brouwer and Fitzpatrick (1998, 2000, 2002 this issue). In those publications it is also shown how distinctions might be made between soil macromorphological features that reflect present hydrological conditions, and those that are a result of past hydrological conditions. The present, second paper deals with detailed piezometric observations and with quantitative relationships between observed soil colours and duration of waterlogging.

Materials and methods

Site description

Soil and hydrological data were collected at Gatum, on the eastern Dundas Tableland in western Victoria, which lies approximately 300 km west of Melbourne (Fig. 1) at 200–300 m above sea level. The Tableland is a flat, dissected surface with deeply weathered regolith. Valley depths are of the order of 20 m, with maximum slopes between 6–8°. Ongoing weathering, possibly since the Permian, has produced deep weathering profiles, with ferricretes and ferruginous gravelly soils overlying mottled and pallid zones that extend to tens of metres depth. Since emplacement, the area has been tectonically stable except for updoming in the order of 240 m (Joyce 1991). Volcanic centres have not been widely recognised, with the exception of the Glendinning Ignimbrite, which contains abundant large locally derived clasts near the base (Simpson 1998).

At Gatum itself the underlying geology is no longer believed to be rhyolite (Spencer-Jones 1965; Sibley 1967) but the Gatum ignimbrites of the Rocklands Volcanic Group that erupted onto the land surface around 410 Ma (Early Devonian) (Cayley and Taylor 1997; Simpson 1997, 1998; Quinn 1997). Depth to fresh basement rock at Gatum varies from more than 20 m to approximately 12 m (Lewis 1985). Above the fresh ignimbrites there is a zone of weathered rock, which, although of low hydraulic conductivity, acts as the main aquifer (see Brouwer and Fitzpatrick 2002, this issue). Because of this low hydraulic conductivity, groundwater hydraulic heads occur on top of the ignimbrites such that groundwater catchment boundaries generally coincide with surface catchment boundaries: the groundwater regime is of the local discharge type (Lewis 1985). This means that local changes in recharge should have an effect on local groundwater levels and salinity, and not be swamped by inflow of groundwater from outside the catchment (Jenkin 1981).

The weathered rock grades upwardly into a pallid zone. In the upper part of the old pallid zone, which can be brilliantly white, there remain only some quartz crystals from the original ignimbrite parent material. It is in this pallid zone that cyclic salts have accumulated over thousands of years, now forming a salt store of the order of 500 tonnes per ha (Brouwer and Van de Graaff 1988). Above the pallid zone there may be a

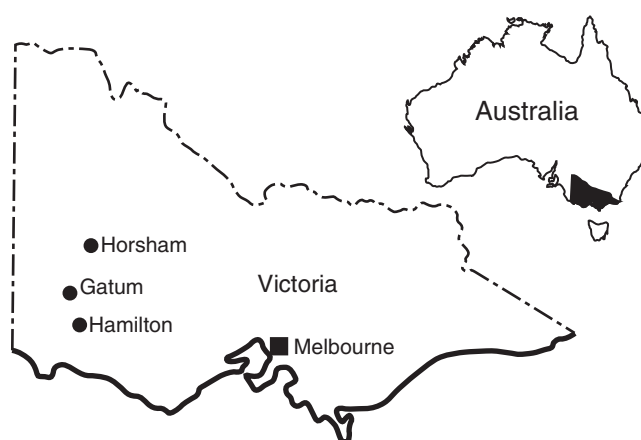


Fig. 1. Site location.

zone of yellow or reddish yellow, relatively coarse sandy clay loam, weakly platy in structure, of unclear origin. Either above this or directly on top of the pallid zone, there is the old mottled zone, strongly and coarsely platy in structure. Of the old laterite hardcaps formerly present above the mottled zone only fragments can still be found, mostly of gravel or stone size, but occasionally with a maximum dimension of more than 0.6 m.

Like similar landscapes with remnants of lateritic paleosols in nearby South Australia (Milnes *et al.* 1985), the Dundas Tableland has probably undergone a number of weathering, erosion, and deposition cycles, and possibly faulting during the uplifting. This has resulted in all of the paleohorizons mentioned outcropping in some part of the landscape, and acting as parent material for soil development in the surface 1 to 1.5 m. The most recent, and still on-going, period of soil formation has resulted in the development of mainly yellow duplex or texture-contrast soils (Gibbons and Downes 1964; Brouwer and Fitzpatrick 2000 and 2002, this issue).

The site at Gatum was selected because of its representativity for geological, hydrological, and land use conditions on the eastern part of the Dundas Tableland. In Brouwer and Fitzpatrick (2000, 2002 this issue), the toposequence studied at Gatum is described in detail. In summary, according to Northcote (1979), the toposequence can be characterised as consisting of hard, pedal, mottled yellow duplex soils, with bleached A2-horizons occurring part way along and at the bottom of the slope. According to Isbell (1996), it can be characterised as consisting of mostly Reticulate Brown Chromosols, with local ironstone gravel concentrations (Ferric), and grading to Hydrosols with salt accumulations (Sodic) part-way along the slope, and at the bottom of the slope. Using Soil Taxonomy (Soil Survey Staff 1996), the toposequence is best described as consisting of Typic Plinthoxeralfs, with Aquic Natrixeralfs occurring part-way along the slope, and Typic Natraqualfs at the bottom of the slope. The occurrence of the Hydrosols or Natrixeralfs part way along the slope is related to the outcropping there of pallid zone material.

The present climate at Gatum is Mediterranean. Annual rainfall averages approximately 630 mm, mostly falling in winter between May and October, when there are also occasional frosts. Average annual pan evaporation is approximately 1400 mm. The vegetative cover before clearing for agriculture and for timber for railway sleepers was red gum (*Eucalyptus camaldulensis*) woodland. This was replaced by pasture, at first using mostly annual pasture species, in the last 15–20 years also perennial species (mainly *Phalaris*). As a result of the tree-clearing, groundwater recharge has increased and the groundwater table has risen at least 10 m, into the pallid zone containing the cyclic salts, resulting in considerable salinity problems, mostly in lower parts of the landscape (Gibbons and Downes 1964; Lewis 1985; Brouwer and Van de Graaff 1988). Nathan (1999) and Dahlhaus *et al.* (2000) convincingly argue that salinity problems already existed before tree-clearing by Europeans but fail to prove that the problem has not become worse in some places on the Dundas Tableland. On the farm where this study took place the farmer was adamant that salinity problems had become worse; piezometer data indicate a 10-m rise in groundwater levels well into the pallid zone with high salt contents (Lewis 1985; Brouwer and Van de Graaff 1988); Nathan and Dahlhaus focus on changes in evapotranspiration rather than the whole water balance (cf. Brouwer and Van de Graaff 1988 and Brouwer 1989, who discuss that an increase in plant water use coupled with a bigger decrease in runoff will lead to increased groundwater recharge); they present no detailed soils data or analysis thereof; and they ignore the role of rootholes in soil hydrology in the area (cf. Brouwer and Van de Graaff 1988; Brouwer and Fitzpatrick 1998; and this paper).

Hydrological observations

Precipitation was measured approximately weekly using a 0.30-m-high plastic manual rainauge with a 150-mm-diameter circular catchment area, placed at 1.2 m height. Pan evaporation was also measured weekly, using a Class A pan with a bird guard, installed to Bureau of Meteorology specifications. During periods when evaporation exceeded precipitation, the water level in the pan was kept at the prescribed level using a water reservoir located 6 m downwind, connected to the pan by a tube and a float valve.

Soil hydrological monitoring took place between January 1987 and November 1988. Observations were mostly made at approximately weekly intervals, but daily or several times a day during a number of wet periods. On a broad crest (i.e. near site 0 in Fig. 2), 2 sets of shallow piezometers were installed in phalaris-based pasture, to monitor saturated soil conditions in the top 3–4 m of the soil. The piezometers were made from 40-mm-diameter PVC pipe. They were installed to depths where, based on soil colour and structure, restrictions to downward movement of water might be expected to be present: to 0.21–0.26 m depth (top of B1-horizon), to 0.57–0.60 m (top of Bt3 horizon, beginning of fine platy structure), to 0.82–1.00 m (top of Bt4-horizon, beginning of coarse platy structure), and to 3.26–3.28 m (top of C3-horizon or pallid zone, one piezometer only); cf. description of profile G-0 in Brouwer and Fitzpatrick (2000, 2002 this issue). An

additional piezometer was installed to the top of the pallid zone at 3.8 m in May 1988. Similar sets of piezometers were installed elsewhere on the broad crest. They are left out of consideration here, as the data they produced added little to the data obtained from the piezometer sets just described.

The 100-mm-diameter holes for piezometers were hand augered. The bottom part of the hole was scratched with a stick with nails to remove any smearing of the sidewalls. The 40-mm PVC pipe was slotted, inserted, and surrounded by filter sand for the bottom 120–500 mm, the depth depending on filter position relative to shallower soil horizons. The surrounding cavity was then backfilled with a 5 : 1 mixture of clay from the hole and sodium bentonite powder, wetted and compacted in the hole with a rod. For the deeper piezometers some of the back filling away from horizon boundaries was done without the bentonite. For piezometers installed on the top of Bt1-horizons no bentonite was used at all. Piezometers to 18 m depth were installed in 1986, using a Gemco drill rig with an 80 mm diameter screw auger.

The soil profiles encountered during installation of these piezometers were the same for all positions, with only some variations in horizon depth and thickness. Horizontal saturated hydraulic conductivity values at the piezometer sites were measured in August 1987—in the B-horizons using piezometer tests, and in the E-horizons using augerhole tests, both as described by Kessler and Oosterbaan (1974). Residual smearing of the sidewalls from the augering may have reduced the measured hydraulic conductivity. However, all test holes were augered in the same way so all measured conductivities should have been affected in the same way. The relative size of the conductivities should therefore not have been affected to any significant degree.

In May 1989 dye tests were run to check for presence or absence of preferential flow paths. This was at a time when the soil was not yet very wet. In each plot a 1 by 1 m metal frame was driven 20–30 mm into the soil. Water containing approximately 8 g/L of rhodamine B-500 was poured intermittently from a bucket over the surface within these frames. Water within the frame was not allowed to pond more than about 10 mm; when this limit had been reached, pouring was discontinued until all the water had infiltrated. The dye used was similar to that of Johnston *et al.* (1983), who used it for comparable purposes. It was realised that rhodamine can be absorbed to clays, but only a qualitative rather than a quantitative assessment of flow paths was required so some absorption did not matter.

To evaluate duration of waterlogging along the toposequence from the broad crest to the valley floor, shallow piezometers were installed to the top of the Bt1-horizons at seven locations (sites 1–7 in Fig. 2). Locations of these piezometers coincided with profile pits G-1 to G-7 (Brouwer and Fitzpatrick 2000, 2002 this issue). These piezometers, too, were monitored at approximately weekly intervals between January 1987 and November 1988, and daily or several times a day during a number of wet periods. Duration of saturation at the bottom of each piezometer was estimated by assuming that saturation commenced midway between the last observation without water in the piezometer and the first subsequent observation with water in the piezometer. Cessation of saturation was estimated similarly. Duration of saturation at 0.10 and 0.20 m below the surface was estimated using a ruler held horizontally at the appropriate depth in each hydrograph.

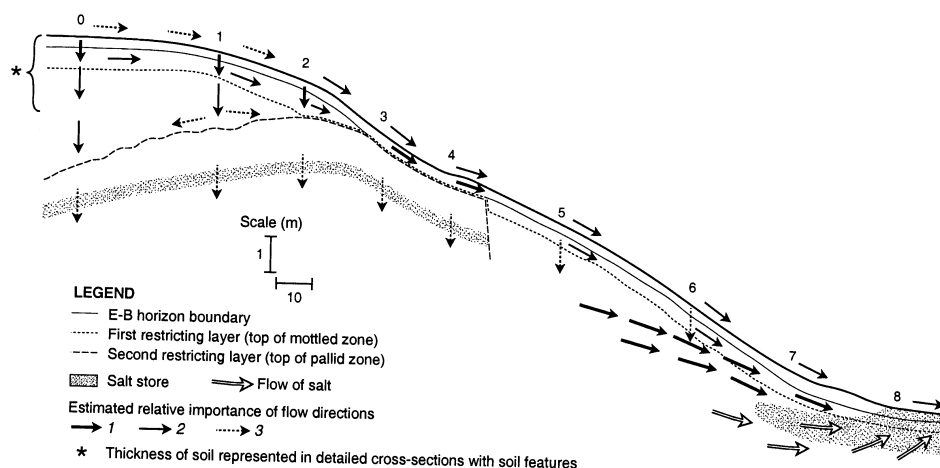


Fig. 2. Hydrological cross-section of the toposequence at Gatam. The numbers indicate piezometer and profile pit positions.

Colour indices

The Bt1-horizons were very variable in thickness, and sometimes too thin for proper calculation of colour indices. Therefore, colour indices were calculated from profile descriptions for Bt2-horizons (0.4–0.6 m depth). For profile G-5, where the Bt2-horizon is only 0.05 m thick, the Bt3-horizon was used. All available information on all 9 profiles can be found in Brouwer and Fitzpatrick (2000). Descriptions of profiles G-0, G-3, G-5, and G-8 are given in Brouwer and Fitzpatrick (2002 this issue). The other 5 profiles are similar to or intermediate between the 4 profiles mentioned. The methods used to calculate colour indices came from 3 sources:

(i) The redness rating RR of Hurst (1977) is defined as:

$$RR = H \cdot V / C$$

where V and C are the (moist) value and chroma of the matrix respectively; H represents the hue of the matrix and is assigned the weight 5 for a hue of 5R, 10 for 10R, 12.5 for 2.5YR, and 20 for 10YR, and proportional weights for the other hues in between.

(ii) Evans and Franzmeier (1988) developed 2 overall colour indices to reflect soil wetness and aeration characteristics: Cc as a Colour index depending solely on soil chromas, Cch depending on soil chromas and hues. For calculating Cc for the soil matrix, the colour index number of the matrix is equal to the chroma number of the matrix: e.g. 6 for 10YR5/6. For hues colour index numbers were assigned as follows: 2.5YR = 17.5, 5YR = 15, 7.5YR = 12.5, 10YR = 10, 2.5Y = 7.5, 5Y = 5.0 N(neutral) = 2.5. Gley colours (hues GY, G, GB, and B) were = 0. Thus for the purposes of calculating Cch, the colour index number of a soil matrix with colour 10YR5/6 is equal to 16, being 6 for the chroma + 10 for the hue.

Mottle colour indices were calculated by Evans and Franzmeier (1988) in a similar way, on the basis of chroma only for Cc, and of chroma + hue for Cch. The mottle colour indices were originally weighted according to their abundance, i.e. the proportion of the ped interior that the mottles covered, each type or colour of mottle separately. The remaining proportion of the ped interior (one minus the sum of the fractions occupied by the various mottles) was assigned as weighting factor to the matrix colour. As the profiles at Gatun were not described with the colour indices of Evans and Franzmeier (1988) in mind, the percentage of the ped interior occupied by all mottles was calculated by taking the mid-value of the mottle abundance class, as noted in the field and expressed as a fraction. Two-thirds of that fraction was assigned to the primary mottles, and one third to the secondary mottles. For example, if abundance of all mottles was 10–20% (mid-value 15%), primary mottles were taken to cover 10% of the total area (weighting factor 0.10), and secondary mottles 5% (weighting factor 0.05).

The colour (chroma or chroma + hue) of argillans was considered to have the same importance as the colour of ped interiors by Evans and Franzmeier (1988). They gave argillan abundance classes the following weighting factors: 0.10 for patchy argillans, 0.50 for discontinuous argillans, and 1.00 for continuous argillans. At Gatun, for clay films or cutans on peds, the size as noted in the field was used instead of Evans and Franzmeier's continuity evaluation. Size classes used were <15 mm (weighting 0.1), 15–30 mm (0.5), and >30 mm (1.0).

To normalise an overall colour index, it is divided by the sum of the weighting factors, i.e. (1 + weighting factor of the argillans). The formula for the overall colour index C of Evans and Franzmeier (1988) thus becomes:

$$C = [(A_m \times C_m + (A_1 \times C_1 + A_2 \times C_2 + \dots A_n \times C_n) + A_t \times C_t) / [1 + A_t]]$$

where A_m = abundance or weighting factor of matrix with colour index number C_m ; $A_m = 1 - (A_1 + A_2 + \dots A_n)$; $A_1, A_2, \dots A_n$ = abundance or weighting factor of mottles with colour index $C_1, C_2, \dots C_n$, respectively; A_t = abundance or weighting factor of argillans with colour index number C_t .

As described above, the colour index number can be based solely on the chroma of the feature concerned (C_c), or on both the chroma and the hue (C_{ch}).

(iii) Mokma and Cremeens (1991) were not completely happy with Evans and Franzmeier's formulae. With increasing duration of saturation Mokma and Cremeens' soil matrices tended to become more neutral in chroma and more yellow in hue, leading to a *decrease* in colour index number and a negative correlation between Cc or Cch and time of saturation. However, in their soils longer saturation was also correlated with the occurrence of mottles with brighter chromas and redder in colour, i.e. an *increase* in colour index number for the mottles. They therefore developed a new colour index, with increasing

index numbers always coinciding with increasing saturation, for the matrix as well as mottles and cutans.

As duration of soil saturation increased the hues of their soils became less red and more yellow, so hues were assigned the following colour index numbers by Mokma and Cremeens: 10R = 0, 2.5YR = 1, 5YR = 2, 7.5YR = 3, 10YR = 4, 2.5Y = 5, 5Y = 6, 0 chroma = 7, and gley page = 8. Matrix chroma decreased as saturation time increased, so the chroma was subtracted from 8, the highest chroma in the Munsell charts. Mottle size was considered more important than mottle abundance, so mottle size classes were assigned the following weighting factors for multiplying mottle colour index numbers with: fine = 0.1, medium = 0.2, coarse = 0.3. Similarly, continuity classes of clay films were given the following weights to multiply clay film colour index numbers with: patchy = 0.1, discontinuous = 0.2, and continuous = 0.3. The colour index of a horizon thus became:

$$CI_h = CI_m + [(S_1 \times CI_1) + (S_2 \times CI_2) + \dots (S_n \times CI_n)] + [C_{cf} \times CI_{cf}]$$

where CI_m is colour index of matrix, S_n is size of mottles with colour index CI_n , and C_{cf} = continuity of clay films with colour index CI_{cf} .

Because the original profile descriptions at Gatum were not made for the purpose of determining colour indices, the mottle size classes used in the calculation of Mokma and Cremeens' (1991) indices were changed to <5, 5–15, and 15–30 mm. In addition, to allow unbiased comparisons all B-horizons were assumed to have 2 colours of mottles. Where only one colour was noted that colour was used for both primary and secondary mottle evaluation. As for the method of Evans and Franzmeier (1988), clay film continuity weighting factors were assigned on the basis of cutan size.

The redness ratings of Torrent *et al.* (1980, 1983) were also tried but could not be used for lack of discriminatory power; most of the Bt2-horizons encountered at Gatum have hues of 10YR, which means they have redness ratings of zero according to Torrent and colleagues.

Results

Rainfall

Monthly on-site precipitation and pan evaporation data for May–October, the wettest time of year, are presented in Table 1. Total pan evaporation during that time of year was the same in 1987 and 1988. Precipitation, however, was considerably greater during the winter of 1988 than in 1987: 501 v. 416 mm. August and September in particular were much wetter in 1988 than in 1987. This is reflected in the various piezometer levels.

Piezometers on the broad crest (sites G-0 and G-1)

Typical hydrographs from 1987 and 1988 for the piezometers on the broad crest (representative of profiles G-0 and G-1) are presented in Fig. 3.

Figure 3a shows that on the crest, the first restrictive layer to downward movement of water is well within the B-horizon, at a depth of 0.8–1.0 m. The piezometers at that depth are the first to hold water and the shallower piezometers only start holding water when the perched water table has risen to their level. The position of this first restrictive layer coincides with the Btv4-horizon, which still has its original coarse (20–50 mm) platy structure, and a grey matrix with mostly red mottles, the mostly unaltered old 'mottled zone' (for details see Brouwer and Fitzpatrick 2000, 2002 this issue). In the remaining part

Table 1. Monthly precipitation (P) and pan evaporation (Eo) at Gatum, winter 1987 and 1988

	May		June		July		Aug.		Sep.		Oct.		Total	
	P	Eo	P	Eo	P	Eo	P	Eo	P	Eo	P	Eo	P	Eo
1987	133	77	67	39	77	51	54	48	28	83	57	129	416	298
1988	103	53	58	35	90	43	86	60	73	77	68	105	501	296

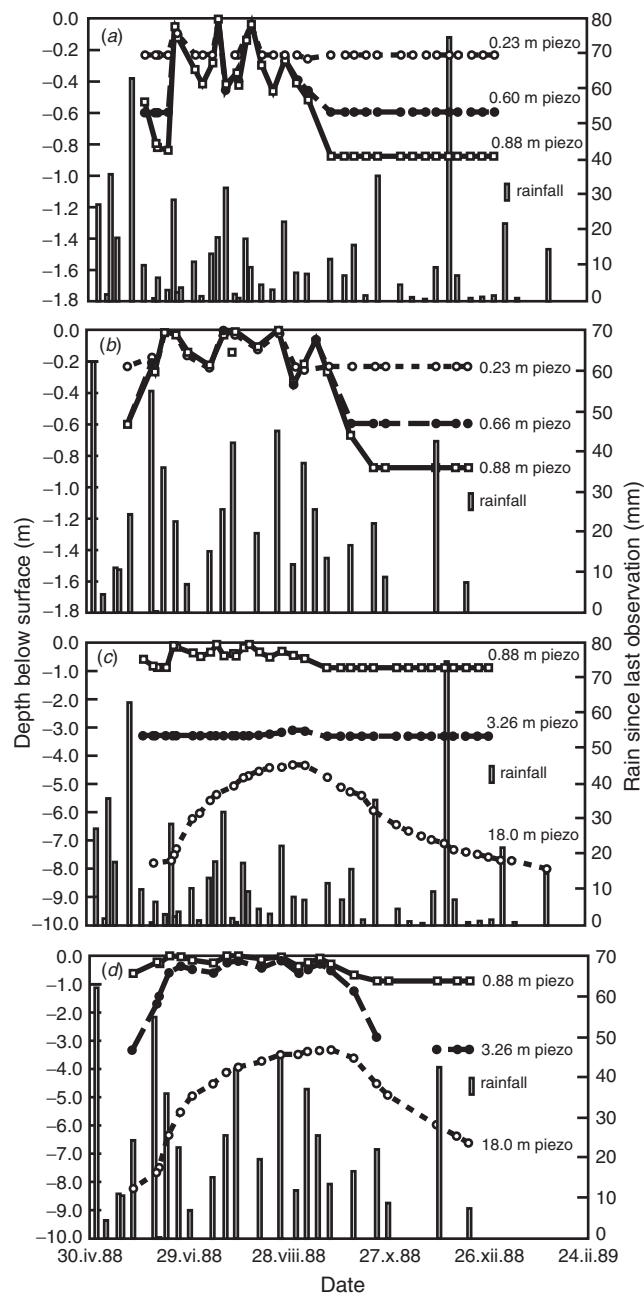


Fig. 3. Piezometer levels on broad crest under conventionally managed phalaris pasture, Gatum: piezometers near the surface in (a) 1987, and (b) 1988; piezometers near the surface and at depth in (c) 1987, and (d) 1988. Horizontal line indicates no water in piezometer.

of this paper it is therefore called 'the restrictive layer at the top of the mottled zone', in contrast to the top of the yellow brown Bt1-horizon. This mottled zone is equivalent to the 'plinthic on crest' subsystem (showing fossil iron enrichment) of the 'lateritic' soil domain in Brouwer and Fitzpatrick (2002 this issue).

In 1988 this restrictive layer at the top of the mostly unaltered mottled zone caused the soil to be saturated to within 0.2 m from the surface continuously from mid-June to mid-

September. Above this restrictive layer cutans have partially changed colour from black to reddish brown, and the mottles have changed from red to more yellow or brown (for details see Brouwer and Fitzpatrick 2000, 2002 this issue). Interestingly, on these soils on the broad crest (including G-0, G-1, and G-2), there is no apparent major restriction to downward movement of water near the top of the Bt1-horizon at 0.2–0.3 m. However, soil morphological data indicate that the restrictive layer may be closer to the top of the Bt-horizon in similar soils well down from the crest; these latter soils are formed in ‘plinthic on slope’ parent material, which probably has a different origin to the ‘plinthic on crest’ material (Brouwer and Fitzpatrick 2002 this issue).

Below the first major restricting layer (Btv4) there is a second one at approximately 2.5–4 m depth, on top of the uneven upper boundary of the C3-horizon, the old ‘pallid zone’ (Fig. 3*b, d*). This restricting layer can cause a second, perched, fresh water table to rise to within 1 m from the surface in winters like in 1988 (Fig. 3*b, d*). This pallid zone is the ‘pallid’ subsystem (showing fossil iron depletion) of the lateritic soil domain in Brouwer and Fitzpatrick (2002 this issue).

In addition there is the permanent, saline (10 000 ppm) water table in the pallid and weathered zones on top of the ignimbrite bedrock at more than 20 m depth. This permanent water table rises to within 3.5 m from the surface in wet winters (Fig. 3*b, d*).

A hydrological cross-section of the near-surface layers along the toposequence at Gatum is shown in Fig. 2. The relative positions of all 3 depths at which water can perch are shown in Fig. 2 of Brouwer and Fitzpatrick (2002 this issue).

Dye-tests and preferential flow paths

The macromorphological observations already indicated that rootholes might be important for downward flow of water at Gatum (Brouwer and Fitzpatrick 2002 this issue). The dye-tests made the role of rootholes abundantly clear. They showed that all major downward movement of water below about 30 mm, and at least as far down as 2.0 m, was along rootholes of all ages, with or without living roots, as well as along some wormholes. This was true for flow through the present yellowish brown B-horizon as well as for flow through the grey- and red-mottled zone. The yellowish brown B-horizon is part of the ‘upper hydromorphic’ soil system of Brouwer and Fitzpatrick (2001, 2002 this issue), which is under the influence of the first perched watertable; the grey- and red-mottled zone, as mentioned, is part of their ‘plinthic on crest’ subsystem, mainly influenced by fossil lateritisation processes. Flow of the dye solution along even major interpedal cracks in both of these soil layers was limited to cracks within about 20 mm from a roothole. The only exception to this was for cracks close to the lower end of a roothole, where greater (up to 50 mm) lateral spread of dye occurred. Many rootholes did not penetrate into the grey- and red-mottled zone. The dye-tests showed that water reaching the restricting layer at the top of that coarse platy grey- and red-mottled zone, at 0.8–1.0 m depth, often could move downward no further. The water had to first flow horizontally to a roothole which did penetrate into the mottled zone, before continuing its downward movement. The dye-tests thus indicate that rootholes are much more important to downward movement of water than inter-pedal cracks. From the macro-morphological studies this was not immediately obvious.

The rootholes were found to extend upward from the top of the grey- and red-mottled zone into the yellowish brown B2- and B1-horizons, and even into the E- and A-horizons. These rootholes tend to have a large flow capacity. As an example, during several hours a

flow equivalent to 115 L/day was measured coming out of single roothole at 1.6 m depth. One such roothole per square meter would already give a saturated soil hydraulic conductivity of 0.12 m/day. In practice rootholes are found much closer together than that, on average about 0.6 m apart or 3 rootholes per square meter in a rectangular spacing (Brouwer and Van de Graaff 1988). However, although the piezometer tests indicate that K_{sat} just above the grey- and red-mottled zone is approximately 0.2 m/day, there is very little lateral gradient; horizontal flow to rootholes is therefore quite slow. As a result, a perched water table often builds up during winter in the yellow brown B-horizon, often reaching the surface during wet periods (Fig. 3a, c). Increasing horizontal conductivity, e.g. through deep ripping, will increase horizontal conductivity and reduce waterlogging. But under pasture at least it will most likely also increase ground water recharge (Brouwer and Van de Graaff 1988).

Saturation along the toposequence and soil colour

The total duration of saturation along the toposequence during 1987 and 1988, at 0.10 and 0.20 m depth, is presented in Fig. 4. The values at point 0, at 0 meters from the broad crest, are derived from the data presented in Fig. 3. Depending on position along the toposequence and on rainfall distribution, saturation commenced in June or July and continued, often intermittently, until September or October. Profile G-8 did not have a piezometer installed but from visual field observations (free water at the surface) was estimated to be saturated to the soil surface 4–6 months of the year. In contrast to situations elsewhere (e.g. George and Conacher 1993), almost all the landscape is waterlogged at least part of the year. A striking feature in Fig. 4 is the relatively long duration of saturation of up to almost 4 months at only 0.2 m below the surface in the midslope profiles G-3 and 4, and to some extent also G-2 (up to two-and-a-half months). This long duration of saturation coincides with the second restricting layer (the pallid zone or pallid soil subsystem), and the second perched water table, coming close to the surface (Fig. 2).

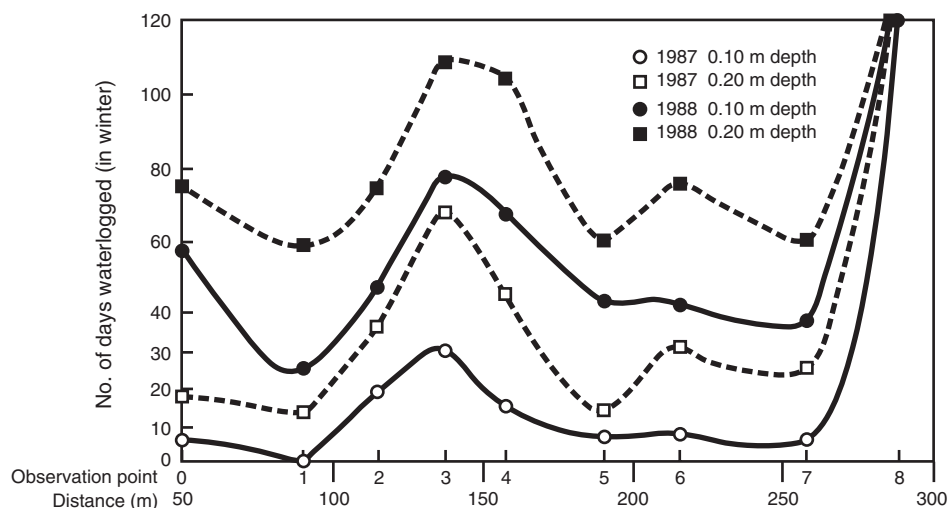


Fig. 4. Duration of saturation in the A- and E-horizons along the toposequence at Gatam.

Table 2. Duration of saturation (average number of days per year), soil colour, and soil colour indices along the toposequence at Gatum

Profile	G-0	G-1	G-2	G-3	G-4	G-5	G-6	G-7	G-8	r^2 linear regression v. saturation
Duration of saturation at bottom of E-horizon	48	30	56	98	98	40	51	78	120	—
Colour of the E-horizon ^A										
Value—moist	4	4	4.5	4.5	4.5	4.5	4.5	5.5	6	0.50
Chroma—moist		4	2	2.5	2	3	2.5	2	2	0.52
Value—dry		5	7	5.5	5.5	6	5.5	7.5	8	0.24
Chroma—dry		4	2	2	2	3	2	2	2	0.49
Colour indices of the B2-horizon ^B										
Value and chroma	6/8		5/5–4/3	5/6–6/4		5/6	5/6		6/3	—
Hurst (1977) redness rating	15		av. 23.3	av. 23.3		16.7	16.7		33.3	0.81
Evans and Franzmeier (1988) Cc (chroma)	5.9		3.5	4.3		2.7	3		3.1	0
Evans and Franzmeier (1988) Cch (chroma + hue)	19.1		15.9	16		18.6	18.5		13.1	0.80
Mokma and Cremeens (1991) matrix (hue)	4		4	4		4	4		4	n.a.
Mokma and Cremeens (1991) matrix (chroma)	0		4	4		3	2		5	0.48
Mokma and Cremeens (1991) matrix (hue + chroma)	4		8	8		7	6		9	0.48
Mokma and Cremeens (1991) matrix + mottles	5.2		9	9.2		7.3	7.2		13.2	0.76
Mokma and Cremeens (1991) matrix + mottles + films	6.8		10.6	12.05		10	9.9		16.2	0.77

^AHue 10YR in all profiles.^BMatrix hue 10YR in all profiles.

Saturation data together with results of different Bt2-horizon colour index calculations are presented in Table 2. The correlation coefficients for linear regressions in Table 2 indicate that the Bt-horizon redness rating of Hurst (1977); the matrix + mottles, and matrix + mottles + clay films indices of Mokma and Cremeens (1991); and the chroma + hue index of Evans and Franzmeier (1988), are more or less equally useful for the prediction of the duration of waterlogging at the bottom of the E-horizon. For all 4 methods the r^2 for linear regression lies between 0.76–0.81. Data on the duration of the saturation of the B2-horizon itself are unfortunately not available. Note that all the E-horizons and all the B2-horizons encountered at Gatum had hues of 10YR.

Discussion

Restricting layers and preferential flow paths

In the past, not finding a restricting layer at the top of Bt-horizons in texture contrast or duplex soils or podzolic soils would have been unexpected. For instance, Stace *et al.* (1968, p. 346) discuss the genesis of lateritic podzolic soils, which grouping includes the soils examined here. Although these authors mention alternating partial saturation and drying of pale A- or A2-horizons as one of the main factors involved in their formation, no mention

is made of saturation of the Bt-horizon. Williams (1983, p. 520) does mention that many 'texture-contrast' soils behave essentially as freely draining profiles. While Williams links such free drainage to characteristics of the Bt-horizon (structure, sodicity, clay mineralogy), he does not discuss possible relationships between colours of A-horizons and permeabilities of subsoil horizons. More recently, however, soil scientists in Australia are realising that, on texture contrast soils, seasonal saturation of the A- and E-horizons does not always mean that there is a restriction to downward flow at the top of the B-horizon, even if the E-horizon is relatively pale (e.g. Brouwer and Van de Graaff 1988; Smettem *et al.* 1991; McFarlane and Cox 1992).

Brouwer and Van de Graaff (1988) presented two hypotheses to explain why a relatively pale E-horizon might overlie a relatively heavy-textured (medium clay) Bt-horizon, as happens on certain soils in the 'plinthic on crest' soil subsystem at Gatum, without there being a restriction to downward water movement near the top of the B. Firstly, it may be that, before clearing early this century of the original red gum woodland, there was never much waterlogging of this soil; following clearing waterlogging increased. However, the matrix of the heavy-textured B-horizon (as opposed to the preferential flowpaths) is relatively impermeable and contains little organic matter for reduction reactions to take place under saturated conditions; the Bt2-horizon therefore retained much of its yellow and brown colouring (e.g. Vepraskas and Wilding 1983a; Richardson and Daniels 1993). In (upper part of) the B1- and the E-horizons permeability and availability of organic matter are greater, resulting in some reduction during waterlogging and a paler colour. In the A-horizon, the large amount of organic matter and less severe reducing conditions prevent the soil matrix from becoming paler. There is support for this hypothesis in the waterlogging and soil colour data from the toposequence, as is discussed below.

A second hypothesis is that the light-textured A-horizon is primarily the result of aeolian (or colluvial) deposition of coarser material on top of a more clayey base (Brouwer and Van de Graaff 1988). The deposited material may have been pale-coloured to start with, and organic matter would subsequently have darkened only its upper part. Certainly the very sharp increase in clay content from 11 to 73% from the E- to the Bt1-horizon in profile G-3 (Brouwer and Fitzpatrick 2001, 2002 this issue) suggests that there at least the 2 horizons are of different origin. It is also possible that the 2 suggested mechanisms act in concert in certain circumstances to produce an E-horizon without a throttle near the top of the B-horizon.

As is to be expected, the uppermost water table, in the first hydromorphic subsystem on top of the mottled zone at 0.8 to 1.0 m depth, is the most reactive of the 3-perched water tables (Fig. 3b, d). The second perched water table, in the second hydromorphic subsystem on top of the pallid zone, shows a less sensitive, slightly delayed response, caused by the retarding effect on downward movement of water of the relatively impermeable mottled zone above it. As the pallid zone or pallid soil system restricts the downward movement of water even further, the reaction of the permanent water table (in the third hydromorphic soil system) to rainfall patterns is slowest of all. Even so, the apparently low storage coefficient and low hydraulic conductivity of the pallid and weathered zones do give rise to big annual fluctuations of the permanent water table.

There are obvious differences between 1987 and 1988 in degree and duration of perching of the water tables on the top of the mottled zone and near the top of the pallid zone (Fig. 3b, d). These can be explained by differences in rainfall patterns between the two years. In 1987 total winter rainfall (May–September) on-site was 415 mm; in 1988 it was 503 mm. In addition, as Table 2 and Fig. 3a indicate, during the winter of 1987 there

were frequent relatively dry periods. During such periods the water table perched on top of the mottled zone at 0.8–1.0 m receded (Fig. 3*a, b*). As a consequence, less water flowed down towards the top of the pallid zone, and the perched water table at about 3.5 m depth hardly formed.

In 1988, however, rainfall exceeded pan evaporation almost every week from mid-May until late August; throughout this period the perched water table on top of the mottled zone remained quite high (Fig. 3*c, d*), and more water flowed down preferential flow paths towards the top of the pallid zone. The pallid zone could not cope with this extra volume of water, and the second perched water table rose accordingly (Fig. 3*d*). Similarly, the permanent water table rose higher, and remained high for longer, in 1988 than in 1987 (Fig. 3*b, d*).

The varying effect on water movement of the restricting layer at the top of the unmottled zone is also due to the nature of that layer; it is not a uniformly less-permeable layer, but a less-permeable layer punctured by quite permeable rootholes. As the rootholes extend far upward, the higher the perched water table, the more water can flow into, and down, the rootholes penetrating into the mottled zone, and hence on to the restricting layer at the top of the pallid zone, causing water to perch there (cf. Fig. 3*b, d*). Even where the rootholes do not continue into the pallid zone itself, they increase the area over which water can infiltrate into the mottled zone, thus increasing downward flow and reducing soil saturation near the surface. Similar functioning of macropores was observed by George and Conacher (1993) in Western Australia.

In addition, the more conductive the saturated layers overlying the restricting layer, the more water flows into the preferential flow paths and on down. This is demonstrated by the fact that, in the winter of 1987, which was not consistently wet, in nearby plots that had been deep-ripped, build-up of the perched water table at 1.0 m was less, but at 3.0 m was more, than in the conventionally tilled plots. Ripping increased horizontal conductivity at 0.47–0.61 m depth from 0.15 to 1.5 m/day, leading to more rapid ‘drainage’ of the first perched water table via rootholes leading downward, and thus to build-up of the second (Brouwer and Van de Graaff 1988). Similarly, during the taking of undisturbed cores in 1988, a 3 m deep augerhole in a deep-ripped plot filled to the surface with water within 24 h of drilling; a hole of the same dimensions in a conventionally tilled plot only had 0.5 m of water in it after the same period. These findings underline the importance of rootholes for downward movement of water at Gatum.

Another factor affecting deep infiltration will be the topography of the restricting layer near the top of the unaltered mottled zone on the broad crests. The topography of the restricting layer will to a certain extent be related to the surface topography. This means that relatively low-lying areas are likely to collect lateral runoff as well as interflow, causing them to remain wet longer, and, all other things being equal, to be areas of preferential groundwater recharge. A similar conclusion was drawn by George (1992) for a catchment on the sandplains of Western Australia. It is probably worthwhile to target such areas when selective ‘drying-out’ of parts of the landscape is being contemplated, e.g. through artificial drainage or through the planting of trees (McFarlane and George 1992; O’Loughlin 1992; McSweeney *et al.* 1994). Trees, however, will only work in the long-term if the wet area where they are to be planted is not itself saline (Stolte *et al.* 1997).

The importance of biopores to downward flow of water at Gatum has been pointed out above. A similar situation, with root channels conducting water into massive to platy, mottled lateritic subsoils, has been described by Vepraskas and Wilding (1983*b*). Passioura (1992) also noted the importance of root channels in duplex soils, in his case for the growth

of new deep roots as much as for water movement. The dye-tests also indicated that infiltration beyond the top of the mottled zone might take place well before the overlying horizons are completely at field capacity, even where there are no clear preferential flow paths extending all the way to the surface. Examination of the 8 inspection pits on the crest showed that below approximately 0.4 m the wormholes all follow pre-existing tree and shrub rootholes. So do virtually all roots of living herbs and grasses. It is therefore concluded that, given a particular rainfall and evaporation pattern, it is the 3-dimensional spacing of the (old) tree and shrub rootholes, together with the conductivity and thickness of the overlying saturated horizons, which determine the effect of the throttle at the 0.8–1.0 m depth at Gatum. In addition, the thickness and storage capacity of the overlying horizons (together with their hydraulic conductivity) help determine when saturation to the surface, followed by runoff, will take place.

The restricting layer near the top of the pallid zone is thought to operate similarly to the throttle near the top of the mottled zone. As reported by Johnston *et al.* (1983), movement of water through the pallid zone is most likely through preferential flow paths, either rootholes or fractures. Both types of preferential flow paths are evident in the pallid zones in some of the profile pits further down the toposequence at Gatum, i.e. pits G-2, G-3 and G-4 (Brouwer and Fitzpatrick 2000, 2002 this issue). In a horizontal plane, the fractures in the pallid zone form 2 polygonal patterns: a finer pattern with polygons of 50–100 (–200) mm diameter, determined by the size of the first-order peds; and a coarser one with polygons of 300–600 mm diameter, mentioned also by Lewis (1985, p. 25). The coarser pattern is probably most important for downward flow of water. Some rootholes have been observed to continue from the mottled zone into the pallid zone, connecting with, and going down, the fractures just mentioned.

Note that Fig. 3d shows the water table perched on top of the pallid zone rising to almost soil surface level during the wet winter of 1988. One explanation is that the hydraulic conductivity of the pallid zone is less than that of the mottled zone, for instance because of the pallid zone being pierced by less rootholes. However, as explained above this may be a matter of apparent conductivity rather than of real conductivity. It is also possible that the horizontal conductivity (and the storage capacity) of the soil layer overlying the pallid zone is less than the horizontal conductivity (and the storage capacity) of the layer overlying the mottled zone. That would mean that it will take the water more time to move to the preferential flowpaths through the pallid zone, causing the water table perched on top of the pallid zone to build up more than the water table perched on top of the mottled zone, even if the hydraulic conductivity of the pallid zone itself is equal to or greater than that of the mottled zone.

Although the reaction of the third water table is slowest of the three, it is not all that slow, and, as mentioned, it is not always preceded by the appearance of the second water table on top of the pallid zone. Or at least not in the places where piezometers were monitored. If water movement through the mottled zone is not too voluminous, the pallid zone may not cause a perched water table to form. Water movement into and through the pallid zone may also be spatially uneven, and for instance influenced by the topography of the top of the pallid zone. It is also possible that there is another process that causes the third water table under the broad crest to rise. It may be that recharge elsewhere in the landscape puts pressure on the third water table. As the third water table is constricted by pallid zone material particularly in lower parts of the landscape, the added pressure could cause it to back up through the weathered ignimbrite to under the broad crest.

Duration of saturation and soil colour

First of all we note that the size of the mottles, deemed important by Mokma and Cremeens (1991), may be less important in situations as at Gatum. At Gatum most mottles in the B-horizon appear to be conversions of the fossil red mottles still present lower down in the profile (see Brouwer and Fitzpatrick 2000, 2002 this issue). In such a case the thickness of the brown rind often found around the red mottles may be a better indication of relative duration of soil saturation than the size of the entire mottle.

Of the four colour indices that give good correlations with duration of saturation across all the different soil systems at Gatum, the redness rating RR of Hurst (1977) is by far the easiest to determine. It involves only the hue and value and chroma of the soil matrix, and ignores mottles and cutans or clay films. The regression line for this colour index is also the one showing the greatest slope (greatest range of index values for the given range in duration of saturation). The redness rating is therefore the colour index most sensitive to differences in duration of saturation. For the E-horizons along the toposequence at Gatum, the average number of days of saturation per year is approximately equal to $4.3 \cdot \text{RR} - 23$ ($r^2 = 0.81$), where the redness rating RR is calculated for the Bt2-horizon. Note that a higher redness rating indicates longer saturation.

Plots of the E-horizon value and chroma against duration of saturation show that these soil parameters, too, can be very useful in estimating duration of saturation of the bottom of the E-horizon at Gatum, although not when used in a linear regression equation as in Table 2. Figure 5 indicates that there are very good negative correlations between chroma C and duration of saturation of the E-horizon, for moist or dry chromas >2 , for up to about 50–55 days of saturation per year (hues of the E-horizons were all 10YR). The average number of days of saturation in this range is approximately equal to $-13.6 \cdot \text{C}_{\text{moist}} + 83$ ($r^2 = 0.98$), or $-10.5 \cdot \text{C}_{\text{dry}} + 72$ ($r^2 = 1.00$). Where moist or dry chromas of the E-horizons were equal to 2 (grey brown and pale brownish grey E-horizons in Fig. 3d of Brouwer and Fitzpatrick 2002 this issue), all that can be said is that average duration of saturation of the bottom of the E horizon was greater than 50–55 days.

However, when the moist or dry chroma is 2, the moist or dry value V of the E-horizon can be used to estimate more precisely the duration of saturation. At Gatum, the average number of days of saturation per year in these soils was equal to $37.7 \cdot \text{V}_{\text{moist}} - 117$ ($r^2 = 0.88$), or $24.5 \cdot \text{V}_{\text{dry}} - 97$ ($r^2 = 0.79$). There was no clear relation between the degree of strong brown mottling in the A- and E-horizons and duration of saturation of those horizons. Such mottling would be expected to not be present in relatively dry horizons (no concentration of iron having taken place), and in very wet horizons (all iron leached out).

It should be noted that these equations relating duration of saturation at Gatum to soil colour aspects ignore the aberrant relationships found for profiles G-3 and G-4. There are good reasons to believe that the colour of the E-horizons of these profiles has not yet come to an equilibrium with increased duration of saturation, following clearing of the original redgum woodland. In addition to the relatively dark colour of the E-horizons in these profiles, they have a much higher proportion of magnetic ironstone gravel than would be expected from the long periods during which they are saturated (see Brouwer and Anderson 2000). Furthermore, there are present in the E horizons of G-3 and G-4 few, fine, distinct, strong brown mottles; in profile G-7, where the E-horizon consists of similar material and is saturated almost as long, such mottles have all been leached out. A similar delayed reaction of soil colour to changes in land use and soil hydrology has been postulated by Franzmeier *et al.* (1983).

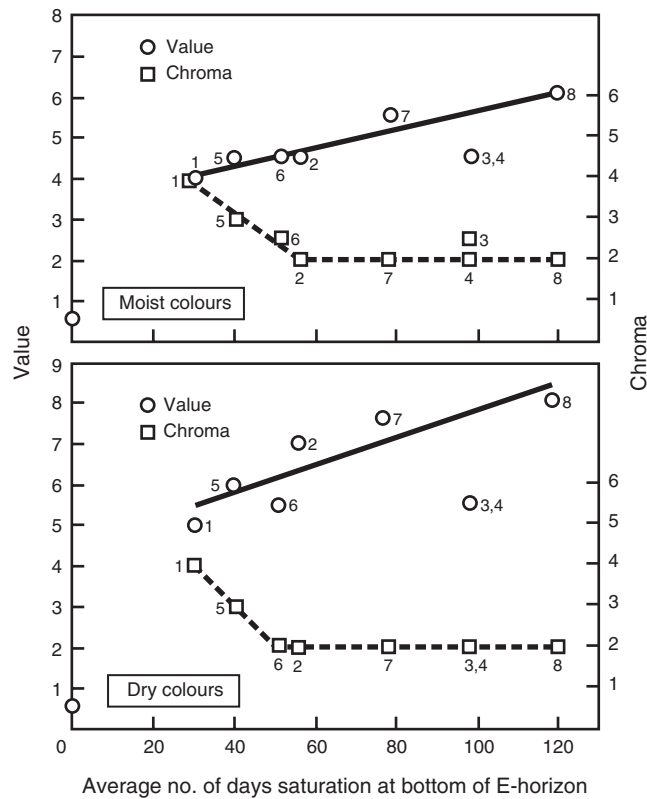


Fig. 5. Duration of saturation v. value and chroma of the E-horizon at Gatun. Numbers indicate position along toposequence.

Conclusions

Based on hydrological observations, the soil hydrological situation on the broad crests at Gatun (with yellow gradational soils or Dermosols) can be summarised as follows. The advantages of the hydrological observations over soil macromorphological observations are indicated where relevant.

(1) There are 3 levels at which downward flow of water is restricted. The first is at the top of the unaltered mottled zone with coarse platy structure, at 0.8–1.0 m depth, well below the top of the B-horizon. During winter this restricting layer gives rise to a fresh perched water table. The second restriction is at the top of the pallid zone, at about 3–3.5 m depth, also giving rise in winter to a fresh perched water table. The third restriction is formed by the unweathered ignimbrite, which underlies the, saline, permanent groundwater table. Soil macromorphological observations had already indicated the likely presence of the two perched water tables, but the piezometer observations and dye studies made it possible to pinpoint the location of in particular the first restrictive layer.

(2) The first perched water table reacts most quickly to the rainfall and evaporation patterns. The permanent water table reacts most slowly. However, because speed of lateral drainage and/or storage capacity decrease with depth, and because the two perched water tables have a possibility to drain downwards that the permanent water table does not have, the permanent water table has the greatest rise and fall over a season, followed, in wet years, by the second perched water table. Soil macromorphological observations obviously cannot provide such information.

(3) Rainfall and evaporation patterns affect the flow of water towards a restricting layer, and thereby affect the extent to which the restricting layer's capacity to conduct water is exceeded and thus the extent to which ponding takes place. If rainfall is well spaced and never excessive, ponding is unlikely to take place on the broad crests at Gatum, as indicated by the hydrographs for 1987, when compared with those for 1988.

(4) The major pathways for downward movement of water below about 30 mm depth, down as far as 2.0 m and sometimes more, are root channels, with or without live roots. Based on macromorphological observations, inter-ped cracks were originally thought to be more important than the dye studies eventually indicated.

(5) Given particular rainfall and evaporation patterns, the 3-dimensional spacing of the rootholes above and through the less-permeable mottled zone, and the conductivity, storage capacity and thickness of overlying horizons, determine the extent of ponding, runoff, and deep infiltration taking place. While this is quite logical, only the use of piezometers showed how important this is.

The following quantitative conclusions could only be drawn thanks to the piezometric observations.

(6) Along the toposequence from crest to valley floor, generally good correlations were found between duration of saturation at the bottom of the E-horizon and colour aspects of the B2-horizons. At the bottom of the E horizons along the toposequence at Gatum, the average number of days of saturation per year is approximately equal to $4.3 \cdot \text{RR} - 23$ ($r^2 = 0.81$), where RR is the redness rating RR of Hurst (1977).

(7) Most previous research has concentrated on the colour of the B-horizon, rather than that of the E-horizon. At Gatum, however, good correlations were also found between the value and chroma of the E-horizon itself and the duration of saturation at its lower boundary. Where the dry or moist chroma of the E-horizon was less than 2, the average number of days of saturation per year was found to equal $-13.6 \cdot \text{C}_{\text{moist}} + 83$ days ($r^2 = 0.98$), or $-10.5 \cdot \text{C}_{\text{dry}} + 72$ days ($r^2 = 1.00$). Where the dry or moist chroma of the E-horizon was 2 or greater, the average number of days of saturation per year in the soils at Gatum was equal to $37.7 \cdot \text{V}_{\text{moist}} - 117$ ($r^2 = 0.88$), or $24.5 \cdot \text{V}_{\text{dry}} - 97$ ($r^2 = 0.79$).

(8) As also indicated by the soil macromorphological data, there is probably significant lateral flow as well as runoff to parts of the landscape that are well above the zone of groundwater discharge, but which are nevertheless relatively low lying, e.g. small swales, or small depressions in the upper surface of restrictive layers which may not be noticeable from surface topography. Such areas are generally saturated longest, and may well be areas of preferential groundwater recharge. They can be targeted for selective drying out through drainage or, if not saline themselves, through tree planting.

(9) The above relationships between soil colour and duration of saturation make it possible to identify areas that are saturated for a particular number of days per year through soil profile inspections during a single visit, instead of through long-term monitoring.

(10) As also indicated by the soil macromorphological data, it is most important to note that, on the broad crest at least, the first effective restriction to downward flow was found at 0.8–1.0 m, and not at 0.2–0.3 m depth, and that rootholes were found to be the major avenue for downward flow. The depth of the first restrictive layer and the role of the rootholes profoundly influence the effect on the water balance of various management options to reduce runoff, waterlogging, deep infiltration or translocation of fertilisers and pesticides. They also affect the economic viability of these management options. It remains to be determined, on texture-contrast soils in Australia, how widespread is the lack of a restriction to water flow at the top of the B, and/or how profound is the influence of

rootholes on soil hydrology. Given the environmental and economic stakes involved, further research into the hydrology and genesis of texture-contrast soils in Australia is of utmost importance.

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